

# Least-squares approach to risk parity in portfolio selection

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The risk parity portfolio selection problem aims to find such portfolios for which the contributions of risk from all assets are equally weighted. Portfolios constructed using the risk parity approach are a compromise between two well-known diversification techniques: minimum variance optimization approach and the equal weighting approach. In this paper, we discuss the problem of finding portfolios that satisfy risk parity over either individual assets or groups of assets. We describe the set of all risk parity solutions by using convex optimization techniques over orthants and we show that this set may contain an exponential number of solutions. We then propose an alternative nonconvex least-squares model whose set of optimal solutions includes all risk parity solutions, and propose a modified formulation which aims at selecting the most desirable risk parity solution according to a given criterion. When general bounds are considered, a risk parity solution may not exist. In this case, the nonconvex least-squares model seeks a feasible portfolio which is as close to risk parity as possible. Furthermore, we propose an alternating linearization framework to solve this nonconvex model. Numerical experiments indicate the effectiveness of our technique in terms of both speed and accuracy.

*Keywords:* Asset allocation; portfolio selection; risk parity; alternating direction method; alternating linearization method

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## 1. Introduction

Quantitative portfolio selection has been a focus of many investors and researchers in the past decades. One central goal in quantitative portfolio selection is to manage the tradeoff between return and risk. Since 1950s, many optimization models have appeared and they continue to play an important role in making investment decisions. However, some strategies resulting from optimal portfolio selection models have been observed to be difficult to implement in practice. One of the commonly cited reasons is the reliance of many of these models on forecasts of future returns. When estimated from historical information, these forecasts can be inaccurate predictors of future behavior of the security returns, and optimal portfolios constructed using these predictors can therefore be inefficient.

One well-known example of an optimal portfolio selection strategy is the mean-variance optimization model. Proposed by Markowitz, mean-variance optimization approach has been regarded as a fundamental framework in portfolio construction. It offered the first quantitative insight into the tradeoff between returns and risk. One persistent criticism of the mean-variance model has been its sensitivity to inputs. Among others, research by Best and Grauer (1991) has shown that slight changes in input parameters, especially in the expected return estimates, can lead to dramatic changes in the optimal portfolio composition.

In practice, some investors apply much simpler asset allocation and portfolio selection strategies and might suffer from their limitations as a consequence. One of these simpler approaches is the well-known “60/40” strategy (60% equity, 40% bonds). From a risk management perspective, the relative balance between equity and fixed income instruments of this strategy is deceptive; because of the higher risk of the equity investments, often more than 90% of the risk of the portfolio comes from equities. Another simple diversification strategy is the “1/n” method, namely, portfolio construction with equal weights in all asset classes; see DeMiguel *et al.* (2009). But, once again, the diversification is achieved only at the capital allocation level and not in terms of risk contributions, simply because the approach does not utilize any information on the assets’ volatility or their correlations.

In this paper, we investigate another portfolio selection strategy – the risk parity approach. The idea of risk parity is not new and can be regarded as a special type of diversification strategy. Using volatility as the measure of risk, the risk parity approach aims to create a portfolio with equal risk contributions from each of the assets in the portfolio. The past few years have witnessed an increasing focus on risk parity research; see, for instance, Chaves *et al.* (2011, 2012), Maillard *et al.* (2010). Most of the existing work considers risk parity in the context of long-only portfolios. In this case, it can be demonstrated that a risk parity portfolio can be obtained from the solution of a convex optimization problem. Under reasonable assumptions, this solution is unique. Some analysis on theoretical properties of long-only risk parity portfolios can be seen in Maillard *et al.* (2010), Spinu (2013).

In the context of long-short portfolios, the risk parity portfolio selection is a richer and more challenging problem. For example, if shorting is allowed multiple risk-parity solutions may exist. We show that a convex model still applies if the investor identifies in advance which assets should be shorted. However, without any such restrictions we show that there could be a combinatorially large number of solutions that satisfy risk parity and these solutions can be identified through an enumeration strategy. In practice, investors may consider adding general bounds on the individual weights of the assets which may limit the total number of risk parity solutions. Moreover, if the bounds are sufficiently tight a risk parity solution satisfying such bounds may not exist at all.

We propose a generalized risk parity model which allows for short sales and applies to cases where risk parity solutions may not exist. Our model is similar to the model proposed in Maillard *et al.* (2010) in that we minimize a function that measures deviation from risk parity in a least-squares sense. However, our formulation has a simpler structure and allows for easier analysis and efficient algorithmic approaches. As in Maillard *et al.* (2010), the optimization model that

we consider is not convex. When they exist, each risk parity solution is a global optimum of the least-squares model. Moreover, no local optimum solution exists on the interior of the feasible set; in other words, all local optima of our proposed formulation occur due to the presence of bounds on the weights. We develop an algorithmic framework based on alternating linearization methods (ALMs) to solve our generalized model. The framework is simple and convergent to a local optimum that is guaranteed to be a global optimum when no constraints are binding. Our formulation also easily extends to the case of the multiple objectives; for example, to the case where one chooses the best risk parity solution according to some additional criterion, when multiple solutions exist.

The rest of the paper is organized as follows. After a brief discussion of the minimum variance optimization framework in Section 2, we introduce the concept of risk parity and consider the convex log-barrier model as well as the proposed least-squares model. In Section 3 we discuss some properties of the local and global optima of our new model. In Section 4, we discuss some extensions based on the least-square model, where we aim to choose the best risk parity solution. We also propose another extension of the risk parity problem to the case where we seek parity on risk contributions of groups of assets rather than individual assets. We propose a class of algorithmic methods based on the ALM framework, to solve the least-square model in Section 5. Experiments and computational results are discussed in Section 6, followed by concluding remarks.

## 2. Risk parity problem

Numerous methods based on the famous Markowitz mean-variance framework have been proposed to overcome its drawbacks while maintaining its advantages; see, for instance, Steinbach (2001). In this paper, we focus on risk based diversification strategies. Unlike the classical mean-variance approach, risk based strategies do not incorporate expected returns into the formulation. Motivations for not using expected returns in the portfolio construction include the difficulty of estimating these quantities accurately, and the well documented sensitivity of the optimal weights to small changes in expected returns. For these reasons, Maillard *et al.* (2010) argue that risk based strategies are more robust than approaches using expected returns.

One prominent example of risk-based strategies in portfolio selection is the minimum variance optimization approach. This approach aims to minimize the volatility of the portfolio, or its active risk, and can be formulated as a convex quadratic optimization problem. This problem can be solved efficiently using widely available optimization software and typically has a unique solution. Here we briefly introduce the minimum variance optimization model to compare it to the risk parity approach.

Suppose we have  $n$  risky assets. Their covariance is given by a symmetric matrix  $\Sigma$  which is assumed to be positive definite. The following optimization problem minimizes the total variance of a fully-invested long-only portfolio:

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^\top \Sigma x \\ \text{s.t.} \quad & x_i \geq 0 \\ & \sum_{i=1}^n x_i = 1, \end{aligned} \tag{1}$$

where  $x = [x_1, x_2, \dots, x_n]^\top$  is the vector of the weights of the  $n$  assets. The factor  $1/2$  in the objective is introduced to simplify the optimality conditions and has no impact on the optimal portfolio. From the first-order optimality conditions of the above problem we see that

$$\Sigma x - \lambda - \gamma e = 0, \tag{2}$$

where  $e$  is an  $n$ -dimensional vector of all ones,  $\lambda \in \mathbb{R}^n$  and  $\gamma \in \mathbb{R}$  are the Lagrange multi-

pliers corresponding to the long-only and full investment constraints, respectively. Note that, complementary slackness conditions imply that if some  $x_i$  is strictly larger than zero, then the corresponding  $\lambda_i$  must be zero. Combining this observation with (2), we obtain

$$(\Sigma x)_i = \gamma, \forall i \text{ s.t. } x_i \neq 0. \quad (3)$$

From (3) we obtain:

$$\left( \frac{\Sigma x}{\sqrt{x^\top \Sigma x}} \right)_i = \left( \frac{\Sigma x}{\sqrt{x^\top \Sigma x}} \right)_j, \forall i, j \text{ s.t. } x_i, x_j \neq 0. \quad (4)$$

In this paper, we make the common choice of using volatility  $\sigma(x) = (x^\top \Sigma x)^{\frac{1}{2}}$  as our risk measure. With this definition of risk, note that  $\frac{\partial \sigma}{\partial x} = \frac{\Sigma x}{\sqrt{x^\top \Sigma x}}$  is the vector of *marginal risk contributions* for the assets in the portfolio. Hence, we have

$$\frac{\partial \sigma}{\partial x_i} = \frac{\partial \sigma}{\partial x_j}, \forall i, j \text{ s.t. } x_i, x_j > 0. \quad (5)$$

The above condition implies that, as long as we invest in an asset, its marginal risk contribution should be the same as that of all other assets with positive weights in the portfolio. As such, minimum variance approach leads to portfolios with equal marginal risk contributions. In practice, while dominating other strategies from the perspective of low volatility, the minimum variance approach often leads to concentrated portfolios, i.e., encourages investors to concentrate on a small number of assets with lower risk profiles and to give up diversification. This behavior is often undesirable and this is exactly what risk parity optimization intends to overcome.

Risk parity portfolios can be motivated by considering Euler decomposition of a portfolio risk measure into contributions from each asset in the portfolio.

**Theorem 2.1:** (*Euler's theorem*) Let a continuous and differentiable function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be a homogeneous function of degree one; i.e. for any constant  $c \in \mathbb{R}$ ,  $f(cx) = c \cdot f(x)$ . Then,

$$f(x) = x_1 \cdot \frac{\partial f}{\partial x_1} + x_2 \cdot \frac{\partial f}{\partial x_2} + \dots + x_n \cdot \frac{\partial f}{\partial x_n}.$$

Simply put, a risk parity portfolio is a portfolio where the total contribution of each asset to the portfolio risk is equal. When  $\sigma(x) = (x^\top \Sigma x)^{\frac{1}{2}}$  is used as the risk measure, using Euler's theorem, we can decompose  $\sigma(x)$  as  $\sigma(x) = \sum_{i=1}^n x_i \cdot \frac{\partial \sigma}{\partial x_i}$ . Here, the quantity  $x_i \cdot \frac{\partial \sigma}{\partial x_i}$  represents the contribution of asset  $i$  to portfolio risk. In this context, the risk parity problem aims to find any portfolio that satisfies

$$x_i \cdot \frac{\partial \sigma}{\partial x_i} = x_j \cdot \frac{\partial \sigma}{\partial x_j}, \forall i, j. \quad (6)$$

We will refer to any solution satisfying (6) as a *risk parity solution*. If we restrict the weight vector to be normalized, we obtain the normalized risk parity problem:

$$\begin{aligned} x_i \cdot \frac{\partial \sigma}{\partial x_i} &= x_j \cdot \frac{\partial \sigma}{\partial x_j}, \forall i, j \\ \sum_{i=1}^n x_i &= 1. \end{aligned} \quad (7)$$

We will refer to any solution of (7) as *normalized risk parity solution*. Note that one can easily convert a solution of (6) into a solution of (7) through simple scaling, as long as the sum of the asset weights is not zero.

We will also consider situations where risk parity solutions may not exist because of the presence of additional restrictions on the portfolio weights. In such cases, we will look for portfolios

that are “close to risk parity” and for this purpose it will be important to quantify the deviation from risk parity. We address this issue in Section 2.3.

Let us assume for a moment that the correlation between any two assets is a constant, that is  $\rho_{ij} = \rho, \forall i, j$ . It can be shown that a closed form solution to (7) can be derived under this assumption; see Maillard *et al.* (2010). This solution is given by the following identity:  $x_i = \frac{\sigma_i^{-1}}{\sum_{j=1}^n \sigma_j^{-1}}$ , where  $\sigma_i^2$  represent the diagonal elements of the covariance matrix  $\Sigma$ . Some literature refers to this case as the “naive risk parity” solution. When the correlations are not constant, a closed-form solution for  $x$  does not exist in general and numerical approaches need to be applied.

## 2.1 Long-only risk parity via convex optimization

Next, we consider the problem of finding a long-only risk parity solution, i.e., finding a vector of weights of  $n$  assets  $x = [x_1, x_2, \dots, x_n]^\top$  such that  $x_i \cdot \frac{\partial \sigma}{\partial x_i} = x_j \cdot \frac{\partial \sigma}{\partial x_j}, \forall i, j$  and  $x \geq 0$ . In this case, it turns out that solving an artificial optimization problem that incorporates a logarithmic barrier term is equivalent to finding a risk parity solution:

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^\top \Sigma x - c \sum_{i=1}^n \ln x_i \\ \text{s.t.} \quad & x_i > 0, \end{aligned} \quad (8)$$

where  $\Sigma$  is the covariance matrix and  $c$  is an arbitrary positive constant. Our use of the logarithmic barrier term in the objective function of (8) is motivated by a related formulation in Maillard *et al.* (2010) that uses a constraint incorporating the sum of the logarithms. Other authors have also utilized the logarithmic barrier function for solving the risk parity problem in independently developed studies; see Kaya and Lee (2012), Spinu (2013).

Since  $\Sigma$  is positive definite and the logarithm function is strictly concave, the objective function of (8) is strictly convex. Under strict convexity, we observe that this convex optimization problem has a unique solution. From the first-order optimality conditions, this solution lies at the point where the gradient of the objective function,  $\Sigma x - cx^{-1}$  is zero, where  $x^{-1} = [1/x_1, 1/x_2, \dots, 1/x_n]^\top$ . Hence, at optimality we have  $(\Sigma x)_i = \frac{c}{x_i}, \forall i$ , which leads to

$$x_i(\Sigma x)_i = x_j(\Sigma x)_j, \forall i, j. \quad (9)$$

It is now easy to see that (9) is equivalent to  $x_i \cdot \frac{\partial \sigma}{\partial x_i} = x_j \cdot \frac{\partial \sigma}{\partial x_j}, \forall i, j$  and that risk parity is achieved at the unique optimal solution of (8).

There is no guarantee that the weights in the solution of (8) will sum to one, so the result may represent a levered or an under-invested portfolio. Fortunately, we have the following result showing the existence and uniqueness of the risk parity solution in the long-only case if we impose the additional constraint that the sum of all weights equals to one. The proof is similar to Theorem 1.1 in Spinu (2013).

**Lemma 2.2:** *Let  $\Sigma$  be a positive definite covariance matrix. Then there exists a unique solution  $x$  which satisfies:*

$$x_i(\Sigma x)_i = x_j(\Sigma x)_j, \forall i, j \quad (10)$$

$$\sum_{i=1}^n x_i = 1, \quad x_i > 0, \quad \forall i. \quad (11)$$

*In fact, any two long-only risk parity solutions differ by a constant factor.*

**Proof:** The objective function of (8) is continuous and strongly convex and it increases to infinity at the boundary of the feasible region. Hence, the intersection of its level set and the

feasible region is compact. This implies that it has a unique global minimum, say,  $x^c$  for any given  $c > 0$ . Consider the following optimization problem:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^\top \Sigma x - \alpha c \sum_{i=1}^n \ln x_i \\ \text{s.t.} \quad & x_i > 0, \end{aligned} \quad (12)$$

where  $\alpha$  is a positive scalar. It is easy to verify that  $\sqrt{\alpha}x^c$  is the unique solution for this problem since  $\sqrt{\alpha}x_i^c [\Sigma(\sqrt{\alpha}x^c)]_i = \alpha c$ . Hence, as  $\alpha$  varies from 0 to  $\infty$ ,  $\sum_{i=1}^n \sqrt{\alpha}x_i^c$  varies from 0 to  $\infty$ . Further, for  $\alpha^* = \frac{1}{(\sum_{k=1}^n x_k^c)^2}$ , the solution of (12), denoted as  $x^*$ , satisfies  $\sum_{i=1}^n x_i^* = 1$ . It is easy to see that  $x_i^* = \frac{x_i^c}{\sum_{k=1}^n x_k^c}$  and  $x^*$  is unique and independent of initial choice of  $c$ .  $\square$

## 2.2 Risk parity solutions over orthants

In this subsection, we explore the set of risk parity solutions when the long-only restriction on weights is removed. The approach discussed in Section 2.1 identifies only the solution in the nonnegative orthant. To find solutions in other orthants, we consider the following modified log-barrier approach:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^\top \Sigma x - c \sum_{i=1}^n \ln \beta_i x_i \\ \text{s.t.} \quad & \beta_i x_i > 0, \end{aligned} \quad (13)$$

where  $\beta = [\beta_1, \beta_2, \dots, \beta_n]^\top \in \{-1, 1\}^n$ , defines the orthant in which the solution is sought. If  $\beta_i = 1$ , then  $x_i \in (0, +\infty)$ ; otherwise,  $x_i \in (-\infty, 0)$ . For each choice of  $\beta$ , (13) is a convex optimization problem and thus it has a unique solution  $x^\beta$ . Since there are  $2^n$  such different  $\beta$  vectors, there are  $2^n$  such solutions. Let  $\bar{\beta} = -\beta$  define the complementary orthant of the orthant defined by  $\beta$ . It is easy to see that  $x^\beta = -x^{\bar{\beta}}$ . We have the following lemma.

**Lemma 2.3:** *Let  $\Sigma$  be a positive definite covariance matrix. Then there exist at most  $2^{n-1}$  solutions which satisfy  $x_i(\Sigma x)_i = x_j(\Sigma x)_j, \forall i, j$ ; and  $\sum_{i=1}^n x_i = 1$ .*

**Proof:** For each  $\beta$  and  $x^\beta$ ,  $\hat{x}^\beta = \frac{x^\beta}{\sum_{i=1}^n x_i^\beta}$  is a normalized risk parity solution satisfying (7) as long as  $\sum_{i=1}^n x_i^\beta \neq 0$ . Note that  $\frac{x_i}{\sum_{i=1}^n x_i} = \frac{-x_i}{\sum_{i=1}^n (-x_i)}$ . Also note that for all unnormalized risk parity solutions in the same orthant either  $\sum_{i=1}^n x_i^\beta = 0$  or  $\sum_{i=1}^n x_i^\beta < 0$  or  $\sum_{i=1}^n x_i^\beta > 0$ . Hence, for every choice of  $\beta$ ,  $\bar{\beta} = -\beta$  generates the same scaled risk parity solution  $\hat{x}^\beta$ , and thus there are at most  $2^{n-1}$  such solutions. In other words, for each orthant, the normalized risk parity solution exists if and only if  $\sum_{i=1}^n x_i^\beta \neq 0$  for any of the unnormalized risk parity solutions in that orthant. If  $\sum_{i=1}^n x_i^\beta < 0$ , then the normalized risk parity solution lies in the complementary orthant.  $\square$

**Remark 1:** Note that a scaled risk parity solution does not exist in the orthant defined by  $\beta$  if  $\sum_{i=1}^n x_i^\beta = 0$  in Lemma 2.3. Hence, there may be fewer than  $2^{n-1}$  normalized solutions. When  $\sum_{i=1}^n x_i^\beta = 0$  the portfolio is “market-neutral”. That is,  $x$  represents a zero investment portfolio and its aggregate exposure to the market is zero. This case is independently interesting and can sometimes be desirable in portfolio selection.

To illustrate Lemma 2.3 with an example, let us consider the simple case when the covariance

matrix is diagonal, i.e.

$$\Sigma = \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \\ & & & \sigma_n^2 \end{bmatrix}.$$

In this case risk parity is equivalent to the following system of  $(n - 1)$  linearly independent equations:

$$\begin{aligned} \beta_1 \sigma_1 x_1 &= \beta_2 \sigma_2 x_2 \\ \beta_1 \sigma_1 x_1 &= \beta_3 \sigma_3 x_3 \\ &\vdots \\ \beta_1 \sigma_1 x_1 &= \beta_n \sigma_n x_n, \end{aligned} \tag{14}$$

where  $\beta_i \in \{-1, 1\}, \forall i \in \{1, \dots, n\}$ . Suppose  $\sigma_i^{-1} + \sum_{j \neq i} \frac{\beta_j}{\beta_i} \sigma_j^{-1} \neq 0, \forall i$ . Then (14), together with  $\sum_{i=1}^n x_i = 1$ , leads to a set of closed form solutions

$$x_i = \frac{\sigma_i^{-1}}{\sigma_i^{-1} + \sum_{j \neq i} \frac{\beta_j}{\beta_i} \sigma_j^{-1}}. \tag{15}$$

Since there are  $2^n$  different  $\beta$ , and each pair  $(\beta, -\beta)$  lead to the same system of equations (14), there are at most  $2^{n-1}$  different solutions. Additionally, in (14), if  $\beta_i = 1, \forall i \in \{1, \dots, n\}$ , then (15) becomes  $x_i = \frac{\sigma_i^{-1}}{\sum_j \sigma_j^{-1}}$ , which is simply the “naive risk parity” solution discussed previously.

Given that there may be exponentially many different risk-parity solutions in the long-short case, additional preferences or restrictions on the weights can be used to narrow down these choices. For example, one may look for a risk parity solution with the least volatility or one with weights lying between given bounds. If investors know a priori which assets are desirable to short, then the log barrier approach of Section 2.1 can be easily extended to find the desired long-short risk-parity portfolio by selecting the appropriate  $\beta$  in (13). In the next two subsections, we discuss efficient methods for finding solutions satisfying additional preferences or restrictions without resorting to enumeration.

### 2.3 Least-squares model with general bounds

The log-barrier approach to finding risk parity solutions does not immediately extend to scenarios with additional constraints or preferences. In particular, when general bounds are added, risk parity solution may not exist, however, up to  $2^{n-1}$  different instances of (13) may have to be solved before infeasibility can be established. Moreover, the log barrier formulation gives no direct guidance on how to produce a feasible solution which may be “close to risk parity”. Simple approaches, such as projecting infeasible normalized risk parity solutions onto the feasible region defined by the constraints may generate portfolios that deviate substantially from risk parity. In addition, it is not clear how to extend this approach to the cases when risk parity is desirable for groups of assets rather than for individual assets (e.g., industry-based groups).

In this section we propose a least-squares formulation for solving the risk parity problem. Our formulation is similar to the following formulation proposed in Maillard *et al.* (2010):

$$\begin{aligned} \min_x \quad & \sum_{i=1, j=1}^n (x_i (\Sigma x)_i - x_j (\Sigma x)_j)^2 \\ \text{s.t.} \quad & a_i \leq x_i \leq b_i \\ & \sum_{i=1}^n x_i = 1. \end{aligned} \tag{16}$$

Above,  $a_i$  and  $b_i$  are arbitrary constants representing the bounds on the weight of the  $i$ -th asset ( $a_i$  can be less than zero if we allow short sales). The objective function of (16) introduces a penalty term for each pair of risk contribution terms  $x_i(\Sigma x)_i$  and  $x_j(\Sigma x)_j$  that are different from each other. Alternatively, one can consider using penalty terms for deviations of risk contributions from their average value:

$$\min_x \sum_{i=1}^n \left( x_i(\Sigma x)_i - \frac{\sum_{j=1}^n x_j(\Sigma x)_j}{n} \right)^2.$$

Our formulation is based on this second objective function, but replaces the average risk contribution term with a free variable  $\theta$  that is also optimized:

$$\begin{aligned} \min_{x, \theta} \quad & \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)^2 \\ \text{s.t.} \quad & a_i \leq x_i \leq b_i \\ & \sum_{i=1}^n x_i = 1, \end{aligned} \tag{17}$$

For future reference, we denote by  $F(x, \theta)$  the objective function of (17):

$$F(x, \theta) := \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)^2. \tag{18}$$

If the optimization problem (17) has an optimal value of zero, then risk parity is achieved. Otherwise, the value of the objective function  $F(x, \theta)$  can be regarded as a minimum variance measure towards our goal.

The two formulations (16) and (17) are equivalent in the sense that any risk parity solution is a solution to both optimization problems. However, our formulation (17) offers a much simpler form of the objective function containing only  $n$  elements in the sum, while the formulation from Maillard *et al.* (2010), contains an order of  $n^2$  elements. Hence, our formulation is computationally less demanding, is easier to analyze and contains fewer nonlinearities. Moreover it allows us to develop efficient optimization approaches as will be seen in Section 5. The auxiliary variable  $\theta$  can always be set to its optimal value based on the following lemma. However, allowing  $\theta$  to be a free variable significantly simplifies the formulation.

**Lemma 2.4:** *For any fixed vector  $x$ , there exists one and only one  $\theta^*$  such that (18) is minimized, and  $\theta^* = (\sum_{i=1}^n x_i(\Sigma x)_i)/n = x^\top \Sigma x/n$ .*

**Proof:** For a fixed vector  $x$ , (18) is a strictly convex unconstrained function of  $\theta$ . Further, the function is minimized when first-order optimality is satisfied, which implies  $\theta^* = (\sum_{i=1}^n x_i(\Sigma x)_i)/n$ .  $\square$

**Example 2.5:** Consider three assets with volatilities  $\sigma_1 = 1, \sigma_2 = 1, \sigma_3 = 2$ , respectively and assume the correlation matrix is the  $3 \times 3$  identity matrix. Thus, the covariance matrix is

$$\Sigma = \begin{bmatrix} 1.0 & & \\ & 1.0 & \\ & & 4.0 \end{bmatrix}.$$

The normalized risk-parity solution in the positive orthant is  $x^0 = [0.4; 0.4; 0.2]$ . Now, suppose we have a restricted feasible region:  $0.5 \leq x_1 \leq \infty, x_2, x_3 \in \mathbb{R}^+$ . If we project  $x^0$  onto the feasible region, we obtain  $x^p = [0.5; 0.35; 0.15]$ . This solution is not “optimal” from the perspective of risk parity solution and the objective function of (17). To see that, consider the solution obtained by

optimizing (17):  $x^{opt} = [0.5; 0.333; 0.167]$ . We see that the objective function value at  $x^p$  equals 0.0143, while at  $x^{opt}$  it equals 0.0128. Hence, in terms of risk parity,  $x^{opt}$  is preferable to  $x^p$ .

We can also compare these two solutions using measures of risk concentration. For this purpose, we consider two metrics: the highest risk contribution by any asset and the Herfindahl index. As its name suggests, the highest risk contribution is defined as

$$HRC(x) := \max_i \frac{x_i(\Sigma x)_i}{x^\top \Sigma x}, \quad (19)$$

while the Herfindahl index is defined as follows:

$$h(x) = \sum_{i=1}^n \left[ \frac{x_i(\Sigma x)_i}{x^\top \Sigma x} \right]^2. \quad (20)$$

The values for the Herfindahl index range between  $\frac{1}{n}$ , for the risk-parity portfolio, and 1, for a perfectly concentrated portfolio.

We observe that  $HRC(x^p) = 0.5405$ , while  $HRC(x^{opt}) = 0.5292$ ; and  $h_{x^p} = 0.4002$ , while  $h_{x^{opt}} = 0.3909$ . While the differences are small, both measures indicate that  $x^{opt}$  is a better solution in terms of risk concentration, suggesting that optimizing (17) leads to less concentrated portfolios.

Note that, unlike problems (1) and (8), (17) is a non-convex optimization problem, hence in theory it is harder to solve and may have local solutions. However, as we will show it is a useful formulation, as simple practical and fast optimization schemes can be developed for this problem. Further, this formulation can be extended to include additional optimization criteria and different variants of risk parity.

### 3. Local and global optima issues

In this section, we investigate conditions under which first-order optimality conditions for (17) are sufficient for global optimality. The Lagrangian function for (17) can be written as

$$\mathcal{L}(x, \theta) = \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)^2 - \lambda_a^\top (x - a) - \lambda_b^\top (b - x) + \gamma \left( \sum_{i=1}^n x_i - 1 \right), \quad (21)$$

where  $\lambda_a, \lambda_b \in \mathbb{R}_+^n, \gamma \in \mathbb{R}$ . We now derive the first-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial F}{\partial x_i} - (\lambda_a)_i + (\lambda_b)_i + \gamma = 0, \quad \forall i, \quad (22a)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial F}{\partial \theta} = 0, \quad (22b)$$

$$\sum_{i=1}^n x_i - 1 = 0, \quad (22c)$$

$$a_i \leq x_i \leq b_i, \quad \forall i, \quad (22d)$$

$$\lambda_a, \lambda_b \geq 0, \quad (22e)$$

$$(\lambda_a)_i(x_i - a_i) = 0, \quad (\lambda_b)_i(b_i - x_i) = 0. \quad (22f)$$

The gradient of  $F$  with respect to  $x$  is

$$\nabla_x F(x, \theta) = 2 \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)(e_i \Sigma_i + (e_i \Sigma_i)^\top)x, \quad (23)$$

where  $e_i \in \mathbb{R}^{n \times 1}$  is the  $i$ th column of the identity and  $\Sigma_i \in \mathbb{R}^{1 \times n}$  is the  $i$ th row of the covariance matrix.

When constraints are excluded from (17) we have the following unconstrained problem:

$$\min_{x, \theta} F(x, \theta) = \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)^2. \quad (24)$$

In this case, the first order conditions are simpler:

$$\nabla_x F = 0, \quad \frac{\partial F}{\partial \theta} = 0.$$

We have the following lemma for this unconstrained case:

**Lemma 3.1:** *A solution pair  $\{x, \theta\}$  is a global optimum of (24) with  $F(x, \theta) = 0$  if and only if  $\nabla_x F = 0$ ,  $\frac{\partial F}{\partial \theta} = 0$ .*

**Proof:** 1) If  $F = 0$ , then  $x_i(\Sigma x)_i - \theta = 0$ ,  $\forall i$ , from which  $\nabla_x F = 2 \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)(e_i \Sigma_i + (e_i \Sigma_i)^\top)x = 0$  holds trivially.

2) If  $\nabla_x F = 0$ , then  $x^\top \nabla_x F = 0$ ; hence,

$$2 \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)x^\top (e_i \Sigma_i + (e_i \Sigma_i)^\top)x = 2 \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)(x_i(\Sigma x)_i + (x_i(\Sigma x)_i)^\top) = 0. \quad (25)$$

Let  $B_i = x_i(\Sigma x)_i$ , note that  $B_i = B_i^\top$ . Then, ignoring the constant factor, we have

$$\sum_{i=1}^n (B_i - \theta)B_i = 0. \quad (26)$$

Applying the second condition  $\frac{\partial F}{\partial \theta} = 0$ , which implies  $\theta = \frac{\sum_{i=1}^n B_i}{n}$ , we have

$$n \sum_{i=1}^n B_i^2 = (\sum_{i=1}^n B_i)^2. \quad (27)$$

On the other hand, from Cauchy-Schwarz inequality we know that  $n \sum_{i=1}^n B_i^2 \geq (\sum_{i=1}^n B_i)^2$ . Furthermore, (27) holds only when  $B_i = B_j$  for all  $i, j \in \{1, \dots, n\}$ . Hence,  $F(x, \theta) = 0$ .  $\square$

In contrast, when constraints are imposed, local optima and local stationary points can occur. The following simple example shows that the inclusion of the full investment constraint, even without additional bounds, can lead to a local stationary point that is not a global solution:

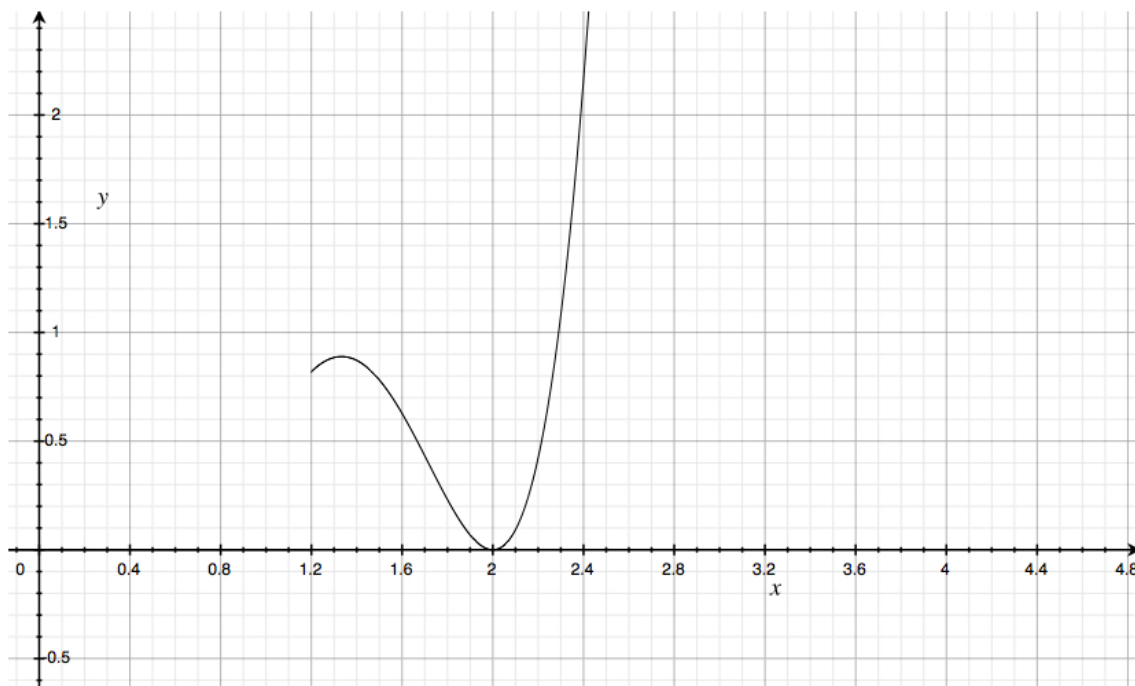


Figure 1. The function value in Example 3.3 with respect to  $x_1$ . Note that, in this  $2 \times 2$  case,  $x_2 = 1 - x_1$ , and  $\theta = \frac{\sigma_1^2 x_1^2 + \sigma_2^2 x_2^2}{2} = \frac{\sigma_1^2 x_1^2 + \sigma_2^2 (1-x_1)^2}{2}$ . Hence, the figure here shows  $F(x_1, x_2, \theta) = (x_1^2 - \frac{x_1^2 + 4(1-x_1)^2}{2})^2 + (4(1-x_1)^2 - \frac{x_1^2 + 4(1-x_1)^2}{2})^2$  when  $x_1 \geq 1.2$ . It shows that  $x_1 = 1.2$  is a local optimum on the boundary.

**Example 3.2:** Consider an example of two assets with volatility  $\sigma_1 = 1, \sigma_2 = 2$ , respectively, and the covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}.$$

Suppose there are no bounds on  $x_1$  and  $x_2$  but we have  $\sum_{i=1}^n x_i = 1$ . Then it is easy to solve the system of equations that satisfies KKT conditions. There are three solutions, two of which,  $x^1 = [\frac{2}{3}, \frac{1}{3}]$  and  $x^2 = [2, -1]$  are risk parity solutions. The third stationary point is  $x^3 = [\frac{4}{3}, -\frac{1}{3}]$ , which is not a risk parity solution. Indeed, this point corresponds to a local maximum of the function  $F$ .

The next example shows that a local minimum can also occur when there are bound constraints in (17), even if there is a risk parity solution, and thus, a global optimum, in the same orthant.

**Example 3.3:** Consider Example 3.2 but with additional bounds  $1.2 \leq x_1 \leq \infty, -\infty \leq x_2 \leq -0.2$ .  $x_1 = 2, x_2 = -1$  is the risk parity solution that satisfies the bounds. As Figure 1 shows,  $x_1 = 1.2, x_2 = -0.2$  is a local minimum but is not a global minimum.

These last two examples suggest that care must be taken in solving (17) if we wish to avoid local solutions and find a global solution of the problem.

#### 4. Extended least-squares models

In this section, we discuss several useful extensions of the risk parity problems that can be easily included in our least-squares model.

#### 4.1 Minimum variance with risk parity

As we discussed above, when short positions are allowed, multiple risk parity solutions may exist. In such cases, the investors have the option to define additional criteria on their portfolio preferences to narrow down the choices for risk parity portfolios, and possibly pick a “best” risk-parity portfolio according to the additional criteria. Introducing preferences based on expected returns is one possible option, but we do not consider this here as we are focused on risk-based strategies. Instead, we seek the risk-parity solution with the least variance. One possible approach to achieve this objective is to enumerate all risk parity solutions and then pick the one with the least variance. Instead of this approach that may require identification of exponentially many risk parity solutions, we consider a multi-objective formulation where the objective function is a weighted sum of total portfolio variance and least-squares risk parity term:

$$\begin{aligned} \min_{x, \theta} \quad & \sum_{i=1}^n (x_i(\Sigma x)_i - \theta)^2 + \rho x^\top \Sigma x \\ \text{s.t.} \quad & a_i \leq x_i \leq b_i \\ & \sum_{i=1}^n x_i = 1, \end{aligned} \tag{28}$$

where  $\rho \geq 0$  is the weight parameter. Note that, in the above formulation we simply added a convex term to the objective function of (17).

Without the risk-parity term in the objective, (28) is simply the minimum variance portfolio problem and is easy to solve using quadratic programming solvers. It is also easy to show that for a large enough  $\rho$  problem (28) is convex in the feasible domain, if this domain is bounded. We propose an approach where we simply solve a sequence of problems (28) with decreasing values of  $\rho$ . This approach is described in Algorithm 1 and is intended to find a risk parity solution with the smallest variance:

**Algorithm 1:** Sequential min-variance risk parity algorithm

1. Choose  $\rho^0 > 0$ ,  $\beta \in (0, 1)$ ,  $\varepsilon > 0$  and  $x^0$ ;
2. for  $k = 0, 1, \dots$ 
  - If  $\rho^k \geq \varepsilon$ , then find  $x^{k+1}$  that solves (28) with  $\rho = \rho^k$  using  $x^k$  as a starting point.  
Set  $\rho^{k+1} := \rho^k \beta$ , CONTINUE.
  - Else, find  $x^{k+1}$  that solves (28)  $\rho = 0$  using  $x^k$  as a starting point. EXIT.

By setting initial  $\rho$  to a large value, we initiate Algorithm 1 with a potentially easy to solve problem and a solution that is close to the minimum variance solution but may be far from risk parity. Then, the algorithm solves a sequence of subproblems of the form (28) with decreasing values  $\rho$ , initializing each new subproblem with the solution of the previous subproblem. The goal is to converge to the risk parity solution that has the smallest variance among all risk parity solutions. Recall that each orthant contains at most one normalized risk parity solution. Algorithm 1 attempts to identify the correct orthant where the minimum variance risk parity solution lies. Hence, once  $\rho$  is small enough (for instance, smaller than some predefined tolerance  $\varepsilon$ , or once the convergence to the correct orthant is apparent), we can drop the minimum variance term and solve the (pure) risk parity problem in the right orthant. In this case, we have a single-orthant risk parity problem as in (13) and can obtain the exact risk parity solution that has small volatility.

Note that, due to the nonconvexity of the objective function, there is no theoretical guarantee that Algorithm 1 will always converge to the minimum variance risk parity solution. However, this approach is significantly more efficient than computing  $2^{n-1}$  possible solutions. In all our experiments, Algorithm 1 reliably produced good solutions and did not get “trapped” in bad local minima.

**Example 4.1:** Often, the long-only risk parity portfolio is conservative in terms of risk, but it may not be the least risky one among all risk parity portfolios, especially in the presence of negatively correlated assets.

Table 1. A comparison of strategies, with the lower and upper bounds to be  $a = -1, b = 2$ .

| Items for comparison      | $x_i$                | $RC_i$              | Volatility |
|---------------------------|----------------------|---------------------|------------|
| Risk parity portfolio (1) | [0.455;0.481;0.064]  | [0.333;0.333;0.333] | 0.289      |
| Risk parity portfolio (2) | [-1.912;1.605;1.307] | [0.333;0.333;0.333] | 3.840      |
| Risk parity portfolio (3) | [1.784;-1.999;1.215] | [0.333;0.333;0.333] | 4.805      |
| Risk parity portfolio (4) | [0.574;0.531;-0.105] | [0.333;0.333;0.333] | 0.238      |

Consider three assets whose volatilities are given by  $\sigma_1 = 1, \sigma_2 = 1, \sigma_3 = 2$ . Moreover, assume the correlation matrix is given by

$$Corr = \begin{bmatrix} 1.0 & -0.9 & 0.3 \\ -0.9 & 1.0 & -0.1 \\ 0.3 & -0.1 & 1.0 \end{bmatrix}.$$

Thus, the covariance matrix is

$$\Sigma = \begin{bmatrix} 1.0 & -0.9 & 0.6 \\ -0.9 & 1.0 & -0.2 \\ 0.6 & -0.2 & 4.0 \end{bmatrix}.$$

Since this is a small instance, it is not hard to empirically check that there are a total of 4 normalized risk parity solutions. We list these solutions in Table 1.

By successively setting the parameter  $\rho$  to be 1000, 10, 0.1, 0.001 and  $10^{-5}$ , and solving the corresponding five subproblems, Algorithm 1 finds the risk parity solution [0.574; 0.531; -0.105]. As indicated in Table 1, this is precisely the minimum variance risk parity solution.

Our results suggest that, our method is very robust with respect to the starting point. More results and larger instances are considered in numerical experiments in Section 6.

It is clear that a similar strategy can be applied to select best risk parity portfolio based on another criterion, such as expected return or value at risk, etc.

## 4.2 Group risk parity

Another interesting extension of the risk parity problem is the case of group risk parity where we seek parity of risk contributions from groups of assets instead of individual assets. This variation is useful in the case when there are a large number of assets; for instance, in the context of equity investing. The assets can be grouped using a common risk factor such as industry membership or market capitalization and we look for equal risk contribution from each factor instead of each individual asset.

One consequence of requiring risk parity at the individual asset level is that each asset receives a positive weight. This is not always desirable when investors prefer to limit the number of positions taken, or when transaction costs are of concern. While it is possible to add cardinality constraints to the formulation to address such concerns, such an approach brings additional computational difficulty. We do not discuss cardinality constraints here in the risk parity context and refer interested readers to Brito and Vicente (2014) for more details. Imposing risk parity at a group level instead of at the individual asset level can also address this issue implicitly and can produce “sparse” solutions with fewer holdings.

For group risk parity, we solve the following nonconvex problem:

$$\begin{aligned} \min_{x, \theta} \quad & \sum_{j=1}^l (\sum_{i \in \mathcal{G}_j} x_i (\Sigma x)_i - \theta)^2 \\ \text{s.t.} \quad & a_i \leq x_i \leq b_i \\ & \sum_{i=1}^n x_i = 1, \end{aligned} \quad (29)$$

where  $\mathcal{G}_j$  stands for the  $j$ th group, and  $l$  is the total number of groups. Here we make two assumptions: 1) we invest in all groups (i.e., we do not aim to pursue group sparsity); 2) there is no overlap (i.e., each asset can only lie in one of the groups). Risk parity between groups is achieved if the optimal value of the objective function is zero. In Section 6, we show how group risk parity may produce a desirable portfolio using sector membership for grouping different assets in an equity portfolio. In the next section we introduce an efficient algorithm that can handle problem (17), as well as the variations (28) and (29).

## 5. Algorithms solving second order least-square problems

In this section, we briefly introduce an algorithm for solving a class of second order least-squares problems, namely the Alternating Linearization Method (ALM). Details of this algorithm and others for this class of problems can be found in Bai and Scheinberg (2013), including convergence analysis and additional numerical results. This algorithm was initially inspired by an alternating linearization algorithm in Goldfarb *et al.* (2013). However, here it is applied to a nonconvex problem and in a substantially different setting.

Consider the following optimization problem:

$$\min_{x \in \mathcal{X}, \theta} F(x) = \sum_i ((A_i x)^\top (B_i x) - \theta)^2, \quad (30)$$

where  $x \in \mathbb{R}^n$ ,  $A_i, B_i \in \mathbb{R}^{m \times n}$  and  $\mathcal{X}$  is a set defined by linear constraints.

Note that the risk parity models we discussed in the previous sections can be embedded into this formulation. The objective function of risk parity problem (17) fits the formulation of (30) by setting  $A_i = \Sigma_i \in \mathbb{R}^{1 \times n}$  as the  $i$ th row of the covariance matrix, and  $B_i = e_i \in \mathbb{R}^{1 \times n}$  as the  $i$ th column of the identity matrix. In case of (29),  $A_j \in \mathbb{R}^{m_j \times n}$  is defined by a submatrix of  $\Sigma$  which correspond to rows with indices from set  $\mathcal{G}_j$ .  $B_j \in \mathbb{R}^{m_j \times n}$  is defined as follows: suppose the  $i$ th row of  $A_j$  corresponds to the  $(k_i)$ th row of  $\Sigma$ , then  $i$ th row of  $B_j$  corresponds to the  $(k_i)$ th row of the identity matrix:

$$(B_j)_{i,k} = \begin{cases} 1, & k = k_i \\ 0, & \text{otherwise.} \end{cases}$$

Note that (30) is equivalent to

$$\min_{x \in \mathcal{X}, \theta} F(x, \theta) = \sum_{i=1}^n F_i(x) = \sum_{i=1}^n (x^\top M_i x - \theta)^2, \quad (31)$$

where  $M_i = A_i^\top B_i \in \mathbb{R}^{n \times n}$ . Clearly  $M_i$  is not generally symmetric or positive semidefinite. Hence (31) is a nonconvex optimization problem. We consider a variable splitting approach which replaces  $F(x, \theta)$  by  $F(x, y, \theta) = \sum_{i=1}^n (x^\top M_i y - \theta)^2$ , s.t.  $y = x$ . For brevity, let us omit  $\theta$  from the variables of  $F$  and use  $F(x, y)$ . Our method generates two sequences  $\{x^k\}$  and  $\{y^k\}$  in such a way that  $x^k \rightarrow x^*$  and/or  $y^k \rightarrow x^*$  where  $x^*$  is a local optimal solution of (30).

Given  $y^k$

$$F(x, y^k) \equiv \sum_{i=1}^n (x^\top M_i y^k - \theta)^2, \quad (32)$$

and given  $x^k$  we have

$$F(x^k, y) \equiv \sum_{i=1}^n ((x^k)^\top M_i y - \theta)^2, \quad (33)$$

Both  $F(x, y^k)$  and  $F(x^k, y)$  are convex functions of  $x$  and  $y$ , respectively, for any given  $y^k$  and  $x^k$ . Let  $\nabla_i F$  denote the partial derivative of  $F$  with respect to either  $x$  ( $i = 1$ ) or  $y$  ( $i = 2$ ). In particular, using the form of (31), we have

$$\begin{aligned} \nabla_1 F(x, y) &= \sum_{i=1}^n 2(x^\top M_i y - \theta) M_i y \\ \nabla_2 F(x, y) &= \sum_{i=1}^n 2(x^\top M_i y - \theta) M_i^\top x. \end{aligned} \quad (34)$$

We now construct the following approximations of  $F(x, y)$ :

$$\begin{aligned} Q_\mu^1(x, y^k) &\triangleq F(x, y^k) + \langle \nabla_2 F(y^k, y^k), x - y^k \rangle + \frac{1}{2\mu} \|x - y^k\|_2^2 \\ Q_\mu^2(x^{k+1}, y) &\triangleq F(x^{k+1}, y) + \langle \nabla_1 F(x^{k+1}, x^{k+1}), y - x^{k+1} \rangle + \frac{1}{2\mu} \|x^{k+1} - y\|_2^2, \end{aligned} \quad (35)$$

where  $\mu$  is some chosen positive scalar. The following is the simple version of our ALM algorithm, shown in Algorithm 2.

**Algorithm 2:** Alternating linearization method (ALM)

1. Choose  $\mu_1^0$ ,  $\mu_2^0$ , and  $x^0 = y^0$ ;
2. for  $k = 0, 1, \dots$  do
  - $x^{k+1} := \arg \min_x Q_{\mu_1^k}^1(x, y^k)$ ;
  - $y^{k+1} := \arg \min_y Q_{\mu_2^k}^2(x^{k+1}, y)$ ;
  - choose new penalty parameters  $\mu_1^{k+1} \in (0, \mu_1^k)$  and  $\mu_2^{k+1} \in (0, \mu_2^k)$ .

In practice, a backtracking strategy can be applied to choose values of parameters  $\mu$  at each iteration. A practical backtracking scheme is shown in Algorithm 3. Note that, in each minimization step we check whether a sufficient reduction has been obtained. If so, the minimization step is accepted and  $\mu$  may be increased, otherwise it is decreased and a new candidate step is computed. Each minimization step in Algorithms 2 and 3 is a solution of a strictly convex quadratic programming problem, which can be obtained efficiently by existing methods and software. We will discuss the implemented method in the next section.

**Algorithm 3:** Alternating linearization method with backtracking (ALM-bktr)

1. Choose  $\mu_1^0$ ,  $\mu_2^0$ ,  $\beta \in (0, 1)$  and  $x^0 = y^0$ ;
2. for  $k = 0, 1, \dots$  do
  - (a)  $x^{k+1} := \arg \min_x Q_{\mu_1^k}^1(x, y^k)$ ;
  - (b) if  $F(x^{k+1}) \leq Q_{\mu_1^k}^1(x^{k+1}, x^k)$  then
    - Choose  $\mu_1^{k+1} \geq \mu_1^k$ ;
  - else
    - find the smallest  $n$  s.t.  $\bar{\mu} := \mu_1^k \beta^n$ ,  $\bar{x} := \arg \min_x Q_{\bar{\mu}}^1(x, y^k)$  and  $F(\bar{x}) \leq Q_{\bar{\mu}_1}^1(\bar{x}, y^k)$ ;
    - $\mu_1^{k+1} := \bar{\mu}/\beta$ ,  $x^{k+1} := \bar{x}$ ;

Table 2. A comparison of strategies, with the lower and upper bounds set to  $a = 0, b = 1$ .

| Items for comparison       | $x_i$                           | $RC_i$                           | Volatility |
|----------------------------|---------------------------------|----------------------------------|------------|
| 1/n rule                   | [0.200;0.200;0.200;0.200;0.200] | [0.119;0.524;0.219;-0.002;0.139] | 7.07%      |
| Minimum variance portfolio | [0.050;0.006;0.000;0.862;0.082] | [0.050;0.006;0.000;0.862;0.082]  | 2.16%      |
| Risk parity portfolio      | [0.125;0.047;0.083;0.613;0.132] | [0.200;0.200;0.200;0.200;0.200]  | 3.04%      |

- (c)  $y^{k+1} := \arg \min_y Q_{\mu_2^k}^2(x^{k+1}, y)$ ;  
 (d) if  $F(y^{k+1}) \leq Q_{\mu_2^k}^2(x^{k+1}, y^{k+1})$  then  
     Choose  $\mu_2^{k+1} \geq \mu_2^k$ ;  
     else  
     find the smallest  $n$  s.t.  $\bar{\mu} := \mu_2^k \beta^n, \bar{y} := \arg \min_y Q_{\bar{\mu}}^2(x^{k+1}, y)$  and  $F(\bar{y}) \leq Q_{\bar{\mu}_2}^2(x^{k+1}, \bar{y})$ ;  
      $\mu_2^{k+1} := \bar{\mu}/\beta, y^{k+1} := \bar{y}$ ;

## 6. Numerical results

### 6.1 A comparison between strategies

We compare several asset allocation strategies on a small data set to illustrate the benefits of the risk parity strategy. Consider the following example with 5 assets and the covariance matrix of percentage returns (returns multiplied by 100) given by:

$$\Sigma = \begin{bmatrix} 94.868 & 33.750 & 12.325 & -1.178 & 8.778 \\ 33.750 & 445.642 & 98.955 & -7.901 & 84.954 \\ 12.325 & 98.955 & 117.265 & 0.503 & 45.184 \\ -1.178 & -7.901 & 0.503 & 5.460 & 1.057 \\ 8.778 & 84.954 & 45.184 & 1.057 & 34.126 \end{bmatrix}.$$

The covariance matrix above suggests that Asset 4 is a low risk asset while Asset 2 is a high risk asset. The other assets are in the middle of the risk spectrum. First, we consider the long-only risk parity case, when the lower and upper bound of the weights are set to be 0 and 1, respectively. As we have proved in Lemma 2.2, in this case, there is a unique risk parity solution.

Table 2 shows a comparison of different strategies in terms of volatility and risk concentration. We compare three strategies: 1/n rule, the minimum variance portfolio, and the risk parity portfolio. The relative risk contribution of asset  $i$ , denoted by  $RC_i$ , is calculated as

$$RC_i := \frac{x_i(\Sigma x)_i}{x^\top \Sigma x}.$$

By construction, the minimum variance strategy has the least volatility among these three strategies. However, the distribution of the risk contributions for the minimum variance portfolio is very skewed — more than 86% of the total risk comes from Asset 4. From Table 2 it can be observed that risk parity strategy also puts a high weight on Asset 4, but has equal risk contribution from each asset by construction. Further, the volatility of the risk parity portfolio is between that of the minimum variance portfolio and the equally weighted portfolio, which indicates that risk parity could be viewed as a compromise between these two strategies.

If there are binding constraints on asset weights a risk parity portfolio may not exist, as discussed in previous sections. For instance, if we change the bounds to be  $a_i = 0.05, b_i = 0.35, i = 1, \dots, 5$ , there is no risk parity solution. Due to its much lower risk profile, Asset 4 requires a much higher weight than other assets to match their risk contributions. Since the 0.35 upper

Table 3. A comparison of strategies, with the lower and upper bounds set to  $a = 0.05, b = 0.35$ .

| Items for comparison       | $x_i$                           | $RC_i$                           | Volatility |
|----------------------------|---------------------------------|----------------------------------|------------|
| 1/n rule                   | [0.200;0.200;0.200;0.200;0.200] | [0.119;0.524;0.219;-0.002;0.139] | 7.07%      |
| Minimum variance portfolio | [0.200;0.050;0.050;0.350;0.350] | [0.280;0.178;0.086;0.034;0.421]  | 4.13%      |
| Optimal parity portfolio   | [0.204;0.060;0.130;0.350;0.256] | [0.256;0.198;0.234;0.027;0.284]  | 4.44%      |

Table 4. A comparison of long-short risk parity solutions by enumeration.

| Items for comparison       | $x_i$                               | Volatility | Orthant         | Complementary orthant |
|----------------------------|-------------------------------------|------------|-----------------|-----------------------|
| Risk parity portfolio (1)  | [0.125;0.047;0.083;0.613;0.132]     | 3.04%      | [+; +; +; +; +] | [-; -; -; -; -]       |
| Risk parity portfolio (2)  | [-0.223;0.074;0.125;0.820;0.204]    | 4.26%      | [-; +; +; +; +] | [+; -; -; -; -]       |
| Risk parity portfolio (3)  | [0.157;-0.128;0.124;0.538;0.309]    | 3.34%      | [+; -; +; +; +] | [-; +; -; -; -]       |
| Risk parity portfolio (4)  | [0.154;0.073;-0.285;0.717;0.340]    | 3.48%      | [+; +; -; +; +] | [-; -; +; -; -]       |
| Risk parity portfolio (5)  | [0.165;-0.118;-0.255;0.537;0.671]   | 3.38%      | [+; -; -; +; +] | [-; +; +; -; -]       |
| Risk parity portfolio (6)  | [-0.700;-0.233;-0.503;3.298;-0.863] | 17.94%     | [-; -; -; +; -] | [+; +; +; -; +]       |
| Risk parity portfolio (7)  | [0.272;0.247;0.450;1.495;-1.464]    | 5.84%      | [+; +; +; +; -] | [-; -; -; -; +]       |
| Risk parity portfolio (8)  | [-0.216;-0.162;0.182;0.726;0.471]   | 4.70%      | [-; -; +; +; +] | [+; +; -; -; -]       |
| Risk parity portfolio (9)  | [-0.245;0.117;-0.393;0.985;0.536]   | 5.01%      | [-; +; -; +; +] | [+; -; +; -; -]       |
| Risk parity portfolio (10) | [0.366;-0.097;-0.201;1.289;-0.356]  | 6.67%      | [+; -; -; +; -] | [-; +; +; -; +]       |
| Risk parity portfolio (11) | [-0.553;0.541;0.934;2.902;-2.824]   | 11.87%     | [-; +; +; +; -] | [+; -; -; -; +]       |
| Risk parity portfolio (12) | [-0.513;0.538;-0.426;2.707;-1.307]  | 11.36%     | [-; +; -; +; -] | [+; -; +; -; +]       |
| Risk parity portfolio (13) | [0.287;-0.103;0.441;1.072;-0.696]   | 5.58%      | [+; -; +; +; -] | [-; +; -; -; +]       |
| Risk parity portfolio (14) | [0.272;0.245;-0.220;1.403;-0.700]   | 5.65%      | [+; +; -; +; -] | [-; -; +; -; +]       |
| Risk parity portfolio (15) | [-0.457;-0.183;0.895;1.979;-1.234]  | 10.87%     | [-; -; +; +; -] | [+; +; -; -; +]       |
| Risk parity portfolio (16) | [-0.208;-0.150;-0.346;0.711;0.992]  | 4.69%      | [-; -; -; +; +] | [+; +; +; -; -]       |

Table 5. A comparison of strategies, without any bounds on asset weights

| Items for comparison       | $x_i$                            | $RC_i$                           | Volatility |
|----------------------------|----------------------------------|----------------------------------|------------|
| 1/n rule                   | [0.200;0.200;0.200;0.200;0.200]  | [0.119;0.524;0.219;-0.002;0.139] | 7.07%      |
| Minimum variance portfolio | [0.050;0.006;-0.012;0.856;0.100] | [0.050;0.006;-0.012;0.856;0.100] | 2.16%      |
| Risk parity portfolio*     | [0.125;0.047;0.083;0.613;0.132]  | [0.200;0.200;0.200;0.200;0.200]  | 3.04%      |

bound prevents this, risk parity cannot be achieved. Instead, we seek approximate risk parity by solving (17).

Let us call the optimal solution to (17) the “optimal parity” solution. From Table 3, we observe that the risk contribution of Asset 4 in the optimal parity portfolio is lower than other assets and the resulting excess is shared roughly evenly among three of the four remaining assets. No asset has risk contribution more than 30% in the optimal parity portfolio.

Next, let us consider the long-short case by removing the bounds on asset weights. In Lemma 2.3, we showed that for the long-short risk parity problem with  $n$  assets, there could be as many as  $2^{n-1}$  solutions. In our example, by solving (13) for all orthants, we obtain  $2^4 = 16$  solutions. We enumerate these solutions in Table 4. We also list the orthant that each solution lies in.

Finally, we compare the long-short minimum variance portfolio to the minimum variance risk parity portfolio. We include the equally weighted portfolio in the comparison, shown in Table 5, for completeness. As discussed in Section 4.1, the minimum variance risk parity portfolio can be found by applying Algorithm 1. The solution, denoted as Risk parity portfolio\* in Table 5, provides a favorable tradeoff between lowering portfolio risk and balancing risk contributions from all assets compared to the other strategies. Further, by examining Table 4, we verify that Risk parity portfolio\* is indeed the minimum variance risk parity portfolio.

Table 6. A comparison of strategies on asset allocation example, with the lower and upper bounds set to  $a = 0, b = 1$ .

| Items for comparison       | Expected excess return | Volatility | Sharpe Ratio | VaR 5% | Highest RC | Herfindahl index |
|----------------------------|------------------------|------------|--------------|--------|------------|------------------|
| 60/40 rule                 | 4.33%                  | 9.36%      | 0.462        | 4.43%  | 95.49%     | 0.8783           |
| 1/n rule                   | 8.59%                  | 12.39%     | 0.693        | 5.80%  | 13.12%     | 0.0900           |
| Minimum variance portfolio | 2.59%                  | 3.55%      | 0.730        | 1.17%  | 61.79%     | 0.5278           |
| Risk parity portfolio      | 6.57%                  | 7.72%      | 0.851        | 3.13%  | 7.14%      | 0.0714           |

## 6.2 Strategic asset allocation

In this subsection, we compare four different static investment strategies using real-world data.

We consider the following common market indices to represent different asset classes: S&P 500, MSCI World (Net), Russell 2500, Russell 2000 Growth, Russell 2000 Value, HFRI Equity Hedged Index, MSCI Emerging Markets (Net), HFRI Emerging MKTS Total, HFRI FoF (Conservative Index), BC Treasury 5-10 Yr, BC US Corporate High Yield Index, JPMorgan GBI-EM Index, JPMorgan EMBI+ Index, S&P Global Natural Resources - Energy Index. Here, we use a monthly sampling frequency, and the sampling period is from Nov. 2002 to Aug. 2012. We use the returns of each one of these asset classes during the sampling period to estimate their annualized expected returns and volatilities. For excess return and Sharpe ratio calculations we estimate the risk-free rate from the 3-month T-bill rate which averages approximately 1.8% annually during the sampling period.

We compare the following four strategies: 60/40 rule (allocated between S&P 500 and BC Treasury 5-10 Yr indices, representing equity and fixed income investments), 1/n rule, the minimum variance portfolio, and the risk parity portfolio. For each strategy, the weights and risk contributions of different asset classes are shown in Figure 2 and 3, respectively. For each one of the resulting portfolios, we compare the expected excess return, volatility, Sharpe ratio, 5% nonparametric Value-at-Risk of these strategies. In addition, we compare two risk concentration metrics, namely the HRC and the Herfindahl index that are defined by (19) and (20), respectively.

It can be observed that for both of 60/40 rule and the equally weighted portfolio, the volatility and 5% VaR are high compared to the minimum variance and risk parity portfolios. The risk parity portfolio achieves the highest Sharpe ratio among the four strategies. In terms of risk concentration, the traditional 60/40 rule is dominated by equity risk (more than 95%), as expected. As in the example of the previous section, the minimum variance portfolio, also has a high risk concentration: almost two thirds of the risk is contributed by the least risky asset, namely the HFRI FoF Conservative Index. This can be clearly observed from Figure 3, showing risk contribution of different assets. Overall, the risk parity portfolio provides a good compromise between balancing risk contributions and achieving reasonable amount of expected returns.

## 6.3 US equity sector allocation

In this section, we consider a sector allocation problem for US equities. We perform a simulation over a 40-year period with annual rebalances and compare the performance of three different strategies. As in Section 6.1, we consider equal weighting, minimum variance and risk parity strategies. Similar experiments or simulated examples can be found in, for instance, Chaves *et al.* (2012), Maillard *et al.* (2010).

We consider 17 sectors representing the US equity universe (food, mines, oil, clths, durbl, chems, cnum, cnstr, steel, fabpr, machn, cars, trans, utils, rtail finan, other) as detailed in Kenneth French's data library.<sup>1</sup> We also use return data provided in this library.

<sup>1</sup>Further details are available from <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/index.html>

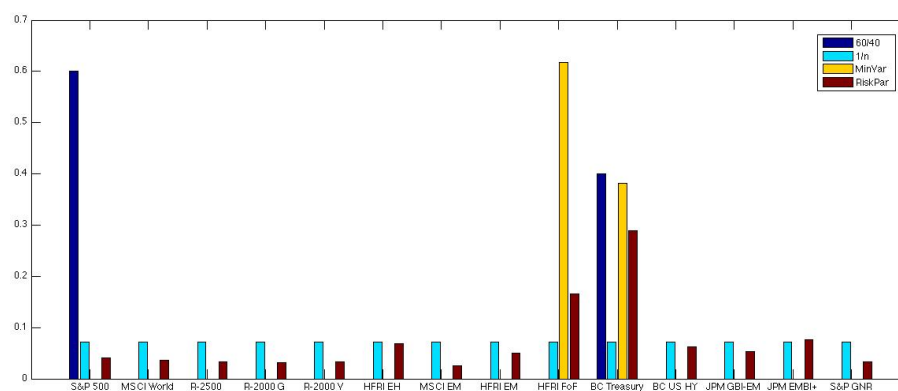


Figure 2. Weights of different asset classes in the asset allocation example

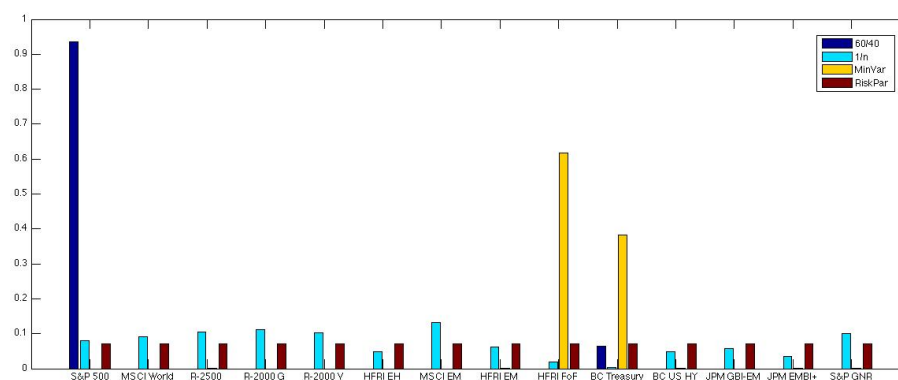


Figure 3. Risk contribution of different asset classes in the asset allocation example

We run simulations using monthly sector returns from Oct. 1972 to Sep. 2012. Since the sampling period is relatively long, we do not assume that input parameters to portfolio construction strategies are constant over time and we take into consideration the time dependency of the return and risk parameters. For each year, we estimate the forward-looking expected returns and covariance matrix based on the monthly returns of the previous  $j$  years, with  $j = 3, 5$  and  $10$  (we do not generate results for the first  $j$  years). The risk-free rate is assumed to be the 3-month T-bill rate. Meanwhile, considering the rebalancing costs, we apply annual rebalancing. The average annualized excess return, volatility, Sharpe ratio and risk contribution are reported. We also plot the excess return and cumulative excess return over time in Figures 4 - 6.

It can be observed from Tables 7, 8 and 9 that minimum variance portfolio has less volatility (and sometimes enjoys a slightly higher Sharpe ratio) than the other two strategies, while its risk concentration is much higher as well. As a comparison, very often the volatility of risk parity portfolio lies between the other two, which once again shows that risk parity is a compromise between equal weighting and minimum variance approaches. Moreover, the risk parity strategy realizes higher returns than the minimum variance strategy under all three moving window lengths considered, as illustrated in Figures 4, 5, and 6.

#### 6.4 Group risk parity portfolios based on the US large cap universe

In this section, we study a stock selection strategy on the US large cap equities as defined by the components of the S&P 500 index. We build our return and covariance estimates using daily returns of S&P 500 constituents, from Aug. 21, 2009 to Aug. 20, 2010 (see:

Table 7. A comparison of strategies for US sector allocation (3 yr rolling windows)

| Items for comparison       | Excess return | Volatility | Sharpe Ratio | Highest RC | Herfindahl index |
|----------------------------|---------------|------------|--------------|------------|------------------|
| 1/n rule                   | 8.04%         | 15.30%     | 0.525        | 9.76%      | 0.0674           |
| Minimum variance portfolio | 6.39%         | 11.67%     | 0.547        | 51.79%     | 0.4089           |
| Risk parity portfolio      | 8.59%         | 13.77%     | 0.624        | 14.72%     | 0.2034           |

Table 8. A comparison of strategies for US sector allocation (5 yr rolling windows)

| Items for comparison       | Excess return | Volatility | Sharpe Ratio | Highest RC | Herfindahl index |
|----------------------------|---------------|------------|--------------|------------|------------------|
| 1/n rule                   | 7.97%         | 15.46%     | 0.516        | 9.83%      | 0.0676           |
| Minimum variance portfolio | 6.65%         | 11.66%     | 0.570        | 49.67%     | 0.3738           |
| Risk parity portfolio      | 7.08%         | 14.11%     | 0.507        | 14.11%     | 0.1329           |

Table 9. A comparison of strategies for US sector allocation (10 yr rolling windows)

| Items for comparison       | Excess return | Volatility | Sharpe Ratio | Highest RC | Herfindahl index |
|----------------------------|---------------|------------|--------------|------------|------------------|
| 1/n rule                   | 8.90%         | 15.15%     | 0.588        | 9.98%      | 0.0684           |
| Minimum variance portfolio | 7.59%         | 11.39%     | 0.667        | 54.90%     | 0.4072           |
| Risk parity portfolio      | 9.57%         | 14.54%     | 0.658        | 16.11%     | 0.1542           |

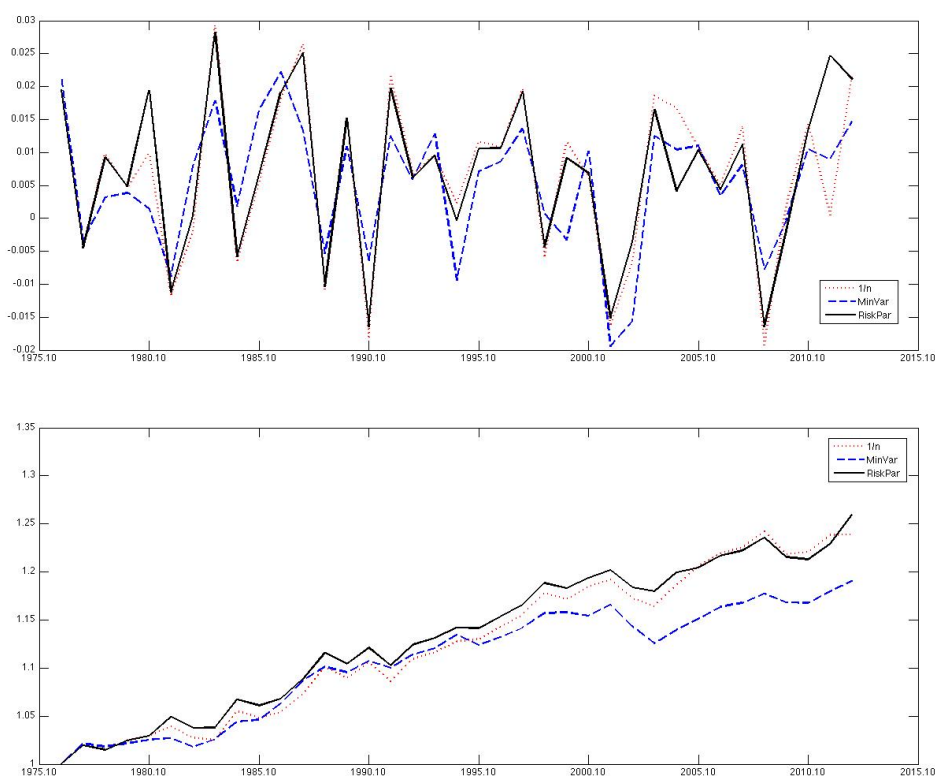


Figure 4. Annual and cumulative returns for US sector allocation strategies (3 yr rolling windows)

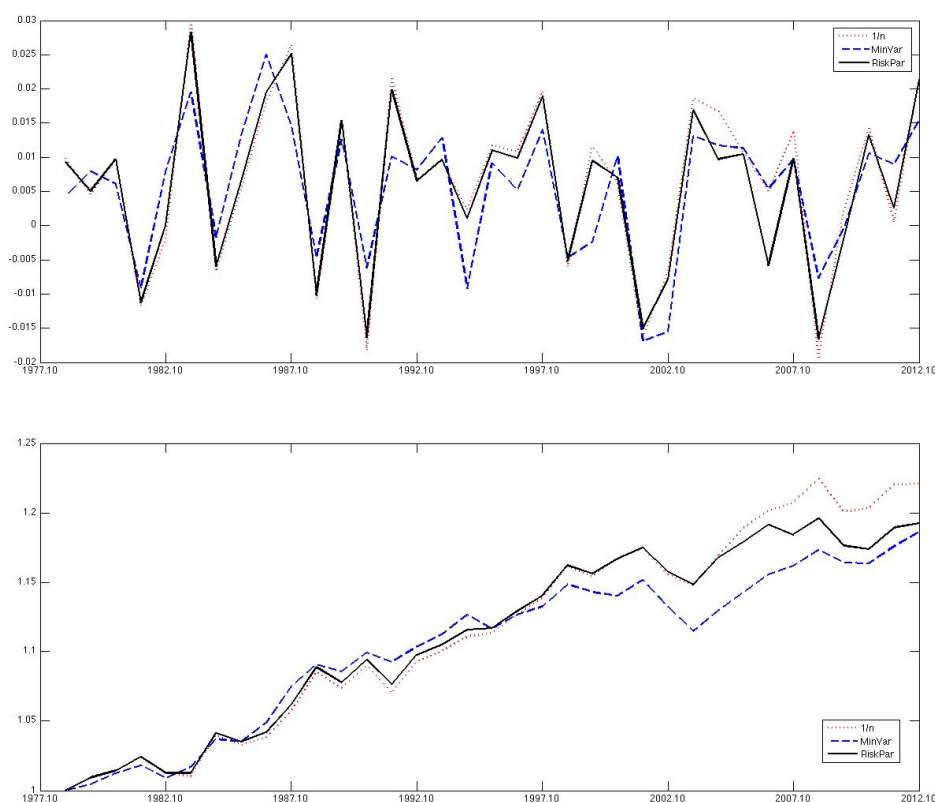


Figure 5. Annual and cumulative returns for US sector allocation strategies (5 yr rolling windows)

<http://pages.swcp.com/stocks/>). We ignore those stocks with less than 200 trading days of data. In total, there are 482 stocks and 245 trading days considered. We now compare the simple “1/n” and minimum variance strategy with group risk parity over this universe.

The groups are determined by Global Industry Classification Standard (GICS Sector) which can be accessed from the site: [http://en.wikipedia.org/wiki/List\\_of\\_S%26P\\_500\\_companies](http://en.wikipedia.org/wiki/List_of_S%26P_500_companies). Using the GICS classification, we partition the S&P 500 stocks into ten sectors: Consumer Discretionary, Consumer Staples, Energy, Financials, Health Care, Industrials, Information Technology, Materials, Telecommunication Services and Utilities. Note that there is no overlap between groups, which means that each stock belongs to one and only one group.

Since the data is sampled from the time period of only one year, we ignore the time dependency of the parameters when testing the performance of different strategies. Here we have 244 daily returns in the sample data. We divide this data into two groups: 163 days for training, and the remaining 81 days for testing. We estimate expected excess returns and covariances using the training data, build a portfolio using various strategies using these estimates, and monitor the performance of the resulting portfolios, out-of-sample, during the testing period.

The results are shown in Table 10. The highest group risk contribution is defined as  $HGRC := \max_j \frac{\sum_{i \in \mathcal{G}_j} x_i (\Sigma x)_i}{x^\top \Sigma x}$ . Also, the group Herfindahl index is defined as

$$h_G = \sum_{j=1}^p \left[ \frac{\sum_{i \in \mathcal{G}_j} x_i (\Sigma x)_i}{x^\top \Sigma x} \right]^2.$$

Since our tests use the validation data, which is different from our training data, we do not

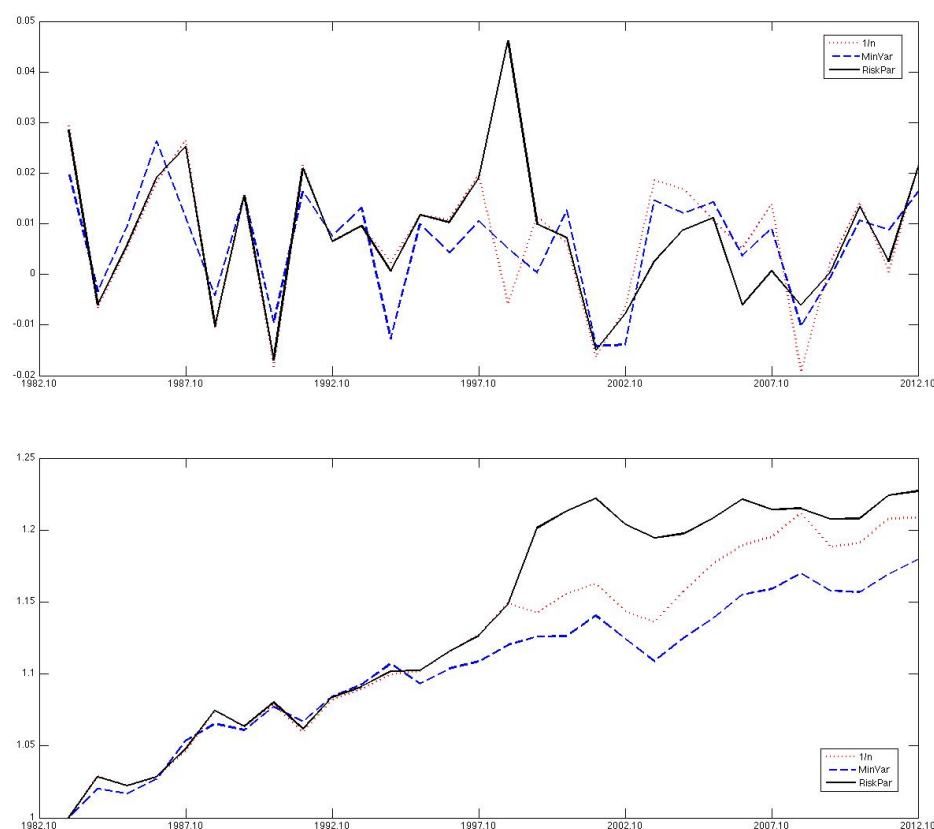


Figure 6. Annual and cumulative returns for US sector allocation strategies (10 yr rolling windows)

Table 10. A comparison of strategies on the US large cap universe, with the lower and upper bounds to be  $a = 0, b = 1$ .

| Items           | Excess return | Volatility | Highest RC | Highest Group RC | Group Herfindahl index |
|-----------------|---------------|------------|------------|------------------|------------------------|
| 1/n rule        | -27.60%       | 28.93%     | 3.94%      | 23.81%           | 0.142                  |
| Min. var.       | -26.03%       | 18.54%     | 40.40%     | 65.10%           | 0.449                  |
| Group risk par. | -19.96%       | 23.46%     | 2.07%      | 11.13%           | 0.101                  |

expect to achieve perfect risk parity during the testing period. But, as can be seen from Table 10, the group Herfindahl index for the group risk parity strategy is not far from the perfect one (which is  $1/10 = 0.1$  since we have 10 groups). As a comparison, the portfolio using 1/n rule has worse excess return (note that in this particular testing period, we have a down market) and higher volatility, and thus is dominated. Moreover, minimum variance portfolio, while having the least volatility, suffers from risk concentration both on stock level and group level. Its highest group risk contribution is 65.10%: more than half of total risk lies in the group “Consumer Staples” (this can be clearly seen in Figure 7). Furthermore, at the individual stock level, the maximum risk contribution is more than 40 percent (General Mills - 40.40%). This is a very high concentration of risk given that the investment universe includes hundreds of securities.

### 6.5 Long-only vs. long-short portfolios

One of the main contributions of this paper is its introduction of strategies for determining long-short risk parity portfolios. In this section, we provide an example to compare long-only portfolios with long-short ones and illustrate that allowing short positions in a risk parity portfolio provides

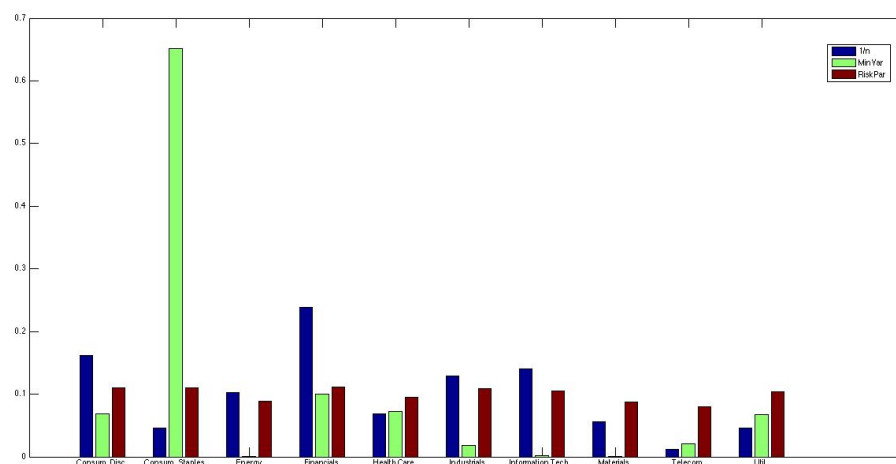


Figure 7. Risk contribution of different groups based on testing data

significant opportunities for reducing volatility and Value-at-Risk while preserving the equal risk contribution property.

Depending on investors' view on the future market trends, three types of long-short portfolios can be considered: they may have a long-side bias, as in the case of popular 130/30 strategies; they may have a short-side bias; or, they may be market-neutral. In our discussion in Section 2, we focused on long-short portfolios with a long-side bias. We proved that risk parity solutions exist in this case and furthermore there could exist as many as  $2^{n-1}$  solutions if we enforce a full-investment constraint.

Risk parity portfolios with a short-side bias can be regarded as complementary counterparts to those with long-side biases and can be found by solving the following problem:

$$\begin{aligned} x_i \cdot \frac{\partial \sigma}{\partial x_i} &= x_j \cdot \frac{\partial \sigma}{\partial x_j}, \forall i, j \\ \sum_{i=1}^n x_i &= -1. \end{aligned} \quad (36)$$

Clearly, a solution of problem (36) can be generated by solving (7) and flipping the sign of each resulting weight.

In contrast, the case of risk parity under market neutrality is more complicated, and it is not clear whether there exists a risk parity solution beyond the trivial one (and, if yes, how many there are). In this paper, we only consider an illustrative asset allocation example with a long-side bias. Market neutral risk parity portfolios will be considered as a subject for future research.

In Table 11 we show a comparison between long-only and long-short strategies on the asset allocation example introduced in Section 6.2. We omit the results for 60/40 rule and 1/n rule since we reported on them earlier. Here, the long-short risk parity portfolio is computed by applying Algorithm 1. The long-short risk parity portfolio has much smaller volatility and VaR when compared to the long-only risk parity portfolio. For comparison, the weights and risk contributions of each asset are shown in Figures 8 and 9, respectively.

It can be observed that, for the long-short minimum variance portfolio, the volatility is less than its long-only counterpart, as expected from the expansion of the feasible region. However, its highest risk contribution and Herfindahl index are also larger, which indicates higher risk concentration. On the other hand, the long-short risk parity portfolio greatly reduces the volatility as compared with the long-only one, without causing any increase in either of the concentration metrics.

Table 11. A comparison of long-only and long-short strategies on asset allocation example

| Items for comparison          | Bounds | Volatility | VaR 5% | Highest RC | Herfindahl index |
|-------------------------------|--------|------------|--------|------------|------------------|
| Long-only min var portfolio   | [0,1]  | 3.55%      | 1.17%  | 61.79%     | 0.5278           |
| Long-short min var portfolio  | [-1,2] | 2.16%      | 1.00%  | 82.93%     | 1.6200           |
| Long-only risk par portfolio  | [0,1]  | 7.72%      | 3.13%  | 7.14%      | 0.0714           |
| Long-short risk par portfolio | [-1,2] | 2.57%      | 1.06%  | 7.14%      | 0.0714           |

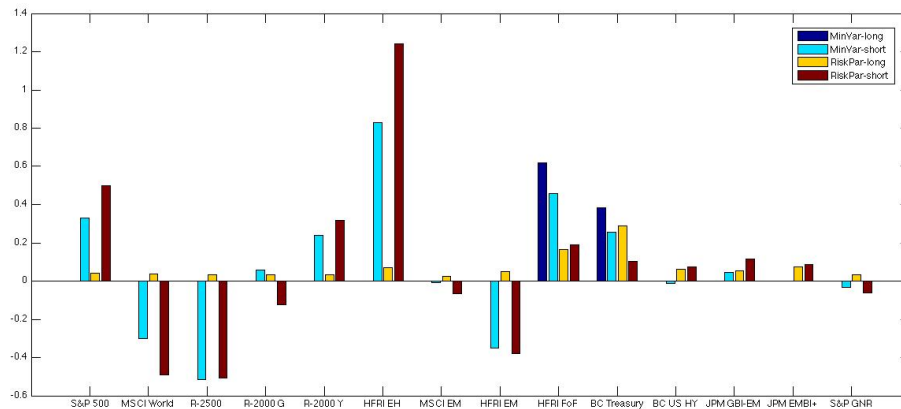


Figure 8. Long-only vs. long-short portfolios: weights of asset classes in asset allocation example

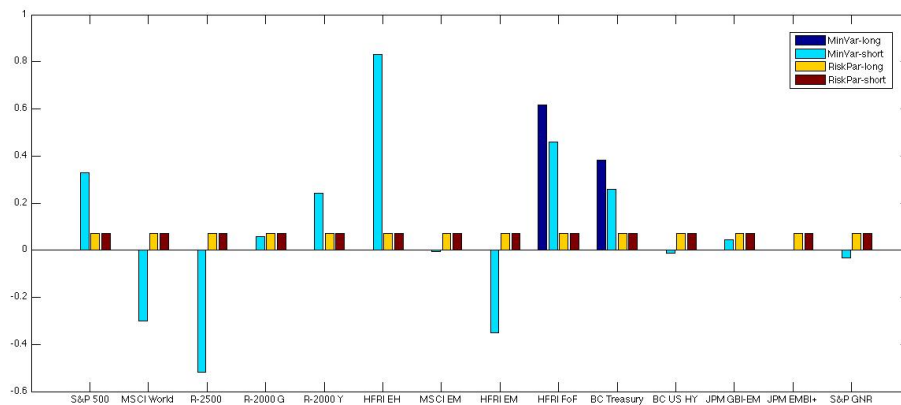


Figure 9. Long-only vs. long-short portfolios: risk contributions of asset classes in asset allocation example

## 6.6 Efficiency of algorithms

While this paper is not aimed at algorithmic details, here we briefly discuss the efficiency of the algorithms shown in Section 5, in application to the instances discussed so far in this section. Further discussion on algorithms and corresponding numerical experiments are provided in Bai and Scheinberg (2013).

Our implementation is written in MATLAB and experiments were performed using MATLAB 2013a on a laptop with Intel Core i5 1.8 GHz CPU and 4GB RAM. We apply Mosek 7.0 to solve the QP subproblems in Algorithm 3, when optimizing  $Q_1$  and  $Q_2$ .

Table 12. A comparison of ALM with backtracking on different instances. The starting point is chosen to be the equally weighted portfolio, i.e.,  $x_i^0 = 1/n$ . Due to the scaling of the data, we chose a large initial  $\mu$ . The number of iterations ( $k$ ), the number of QPs solved, and the objective function value (F-value) are compared.  $\varepsilon$  is the threshold for the largest KKT violation.

| Instance (size)       | Starting $\mu$ | Bounds               | $k$ ( $\varepsilon = 10^{-4}$ ) | QP | F-value                |
|-----------------------|----------------|----------------------|---------------------------------|----|------------------------|
| Random I (20)         | 0.01           | $a = 0; b = 1$       | 6                               | 14 | $1.01 \times 10^{-11}$ |
| Random II (200)       | 0.0001         | $a = 0; b = 1$       | 5                               | 12 | $1.59 \times 10^{-12}$ |
| 5 assets example (5)  | 1000           | $a = 0; b = 1$       | 1                               | 6  | $1.21 \times 10^{-6}$  |
| 5 assets example (5)  | 1000           | $a = 0.05; b = 0.35$ | 1                               | 6  | $1.21 \times 10^{-6}$  |
| 5 assets example (5)  | 1000           | $a = -1; b = 2$      | 1                               | 6  | $1.21 \times 10^{-6}$  |
| Asset allocation (14) | 10000          | $a = 0; b = 1$       | 1                               | 2  | $2.68 \times 10^{-8}$  |
| US equity (482)       | 1000           | $a = 0; b = 1$       | 1                               | 2  | $1.11 \times 10^{-11}$ |

| Instance (size)       | Starting $\mu$ | Bounds               | $k$ ( $\varepsilon = 10^{-6}$ ) | QP | F-value                |
|-----------------------|----------------|----------------------|---------------------------------|----|------------------------|
| Random I (20)         | 0.01           | $a = 0; b = 1$       | 7                               | 17 | $8.86 \times 10^{-15}$ |
| Random II (200)       | 0.0001         | $a = 0; b = 1$       | 7                               | 18 | $6.58 \times 10^{-18}$ |
| 5 assets example (5)  | 1000           | $a = 0; b = 1$       | 11                              | 16 | $1.01 \times 10^{-9}$  |
| 5 assets example (5)  | 1000           | $a = 0.05; b = 0.35$ | 6                               | 20 | $1.63 \times 10^{-7}$  |
| 5 assets example (5)  | 1000           | $a = -1; b = 2$      | 11                              | 16 | $1.01 \times 10^{-9}$  |
| Asset allocation (14) | 10000          | $a = 0; b = 1$       | 6                               | 12 | $3.89 \times 10^{-9}$  |
| US equity (482)       | 1000           | $a = 0; b = 1$       | 2                               | 4  | $1.10 \times 10^{-11}$ |

Table 13. A comparison of algorithms on the 14-asset strategic asset allocation instance in Section 6.2. The starting point is chosen to be the equally weighted portfolio, i.e.,  $x_i^0 = 1/n$ .

| Algorithm           | ALM                                         | $fmincon$ - SQP                             | $fmincon$ - interior point                  |
|---------------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| $x_1 \sim x_7$      | [0.043 0.038 0.035 0.033 0.035 0.069 0.027] | [0.071 0.071 0.071 0.071 0.071 0.071 0.071] | [0.057 0.052 0.047 0.045 0.047 0.090 0.038] |
| $x_8 \sim x_{14}$   | [0.052 0.170 0.259 0.066 0.056 0.083 0.035] | [0.071 0.071 0.071 0.071 0.071 0.071 0.071] | [0.069 0.122 0.116 0.087 0.074 0.109 0.047] |
| $RC_1 \sim RC_7$    | [0.072 0.073 0.073 0.073 0.073 0.070 0.075] | [0.081 0.092 0.104 0.111 0.104 0.048 0.131] | [0.078 0.081 0.082 0.084 0.082 0.074 0.086] |
| $RC_8 \sim RC_{14}$ | [0.072 0.071 0.057 0.072 0.073 0.074 0.073] | [0.063 0.019 0.004 0.049 0.057 0.036 0.101] | [0.075 0.041 0.013 0.076 0.076 0.072 0.081] |
| Success/Failure     | Success                                     | Failure                                     | Partial success                             |
| CPU time (s)        | 0.047                                       | -                                           | 0.457                                       |

In Table 12, we compare the efficiency of the proposed algorithm on different data sets. A random symmetric positive definite matrix can be generated from  $\Sigma = AA^T$ , where  $A_{ij}$  is uniformly distributed within the interval  $[0, 1]$ . Recall that the main computational cost of each algorithm lies in the number of quadratic models it solves as the subproblem. Hence, here we report the number of iterations and the total number of QPs solved. We compare the performance of the algorithms by recording the number of iterations they took until the largest KKT violation (in absolute value) fell below  $10^{-4}$  and  $10^{-6}$ . It can be seen that, after the largest KKT violation is less than  $10^{-6}$ , in most cases the function value is less than  $10^{-9}$ , which indicates that an approximate risk-parity solution has been obtained. In the scenario with tighter bounds, the objective function does not fall below  $10^{-7}$  because there is no risk parity solution.

As an alternative to ALM, optimization methods from MATLAB optimization toolbox can be used to solve (17). However, our results indicate that these methods are unreliable. For instance, for the 14-asset strategic asset allocation example from Section 6.2, we can solve the generalized risk parity problem (17) using MATLAB function  $fmincon$  using two algorithms - SQP and Interior Point. We set the function tolerance (which controls both the size of the latest change in the objective function value and the first-order optimality measure) as  $10^{-8}$  and the maximum function evaluation number as 10000. Table 13 compares the results generated by our ALM with the results obtained by MATLAB's  $fmincon$ . We observe that, in this example,  $fmincon$ -SQP completely fails, as it does not progress from the starting point.  $fmincon$ -interior point performs

Table 14. A CPU time comparison (in seconds) of ALM and Spinu's Newton method for solving instances of various sizes. The starting point is chosen to be the equally weighted portfolio, i.e.,  $x_i^0 = 1/n$ .

| Thresholds         | $10^{-3}$ |        | $10^{-6}$ |        | $10^{-9}$ |        |
|--------------------|-----------|--------|-----------|--------|-----------|--------|
| Instance size      | ALM       | Newton | ALM       | Newton | ALM       | Newton |
| $5 \times 5$       | 0.0146    | 0.0005 | 0.0188    | 0.0005 | 0.0210    | 0.0005 |
| $10 \times 10$     | 0.0188    | 0.0006 | 0.0188    | 0.0006 | 0.0213    | 0.0006 |
| $20 \times 20$     | 0.0185    | 0.0008 | 0.0185    | 0.0008 | 0.0214    | 0.0008 |
| $200 \times 200$   | 0.2597    | 0.0275 | 0.2597    | 0.0275 | 0.2597    | 0.0275 |
| $1000 \times 1000$ | 6.4051    | 2.2817 | 8.1324    | 2.2817 | 8.1324    | 2.2817 |

better but still not nearly as well as our algorithm — it takes longer to find a far worse solution than ALM. Our ALM algorithm finds a solution which is close to optimum by taking only 12 iterations and by solving 26 QPs. This result shows the efficiency of the algorithm we proposed.

We also compare ALM with other methods to solve the risk parity optimization problem. In particular, Spinu (2013) applies a damped Newton method to solve the long-only risk parity problem (8) — the log-barrier risk parity model. Spinu (2013) shows that the log-barrier function is self-concordant and (8) can be efficiently solved using convex optimization. In contrast, our generalized method solves (17) — a nonconvex quartic which is potentially more difficult. Here we test both methods on randomly generated covariance matrices with different sizes. Table 14 compares the efficiency in terms of CPU time required to reduce the chosen deviation measure below a certain threshold value. Here, we choose the deviation measure to be  $\sum_{i=1}^n (\frac{x_i(\Sigma x)_i}{x^T \Sigma x} - \frac{1}{n})^2$ , and thresholds to be  $10^{-3}$ ,  $10^{-6}$  and  $10^{-9}$ . Note that, for both methods, the corresponding optimal solution has a deviation measure zero when no bounds are enforced. As shown in Table 13, Newton method is faster than the ALM approach in all the instances we tested. This is not surprising since, seeking to solve a more general problem, our model is computationally more complex. Further, it does not utilize the full second-order information, as shown in Bai and Scheinberg (2013), because we want to ensure that subproblems are convex. As seen in Section 2.3, the log-barrier approach does not directly extend to the cases when general bounds are added and thus we here only compare the long-only full investment case. These results indicate that, when no customized bounds are needed, the convex model is more efficient but its application is limited compared to our general framework, which is reliable and comparatively efficient as well.

## 7. Conclusion and future work

In this paper, we discuss the problem of finding portfolios that satisfy risk parity of either individual assets or groups of assets as closely as possible. We analyze the limitations of the convex optimization approach, which was proposed in prior literature. We then propose an alternative nonconvex least-squares model whose set of optimal solutions includes all risk parity solutions. We also propose a modified formulation which aims at selecting the most desirable risk parity solution according to some predetermined criteria. Our model has many advantages, especially when general bounds are considered or when other constraints are added. Furthermore, we propose an alternating linearization framework to solve this nonconvex model. Numerical experiments indicate the effectiveness of our technique. Extensions of our models and methods to broader domains remain a topic of future research.

## Supplemental data

The return and covariance data for Section 6.2 can be found at the first author's website: <http://phd.ie.lehigh.edu/~xib210/>. Supplemental data for Section 6.3 can be accessed at: [http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

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